

Open Design Project 5: Functional Prototype

Design Documentation

Parts

- Tight-Seal Strip Brushes
 - Specification:
 - (Length = 1ft) Tight-Seal Strip Brush with Nylon Bristles, 5/16" Wide x 5/16" High Backing, 1" Overall Height
 - Fabrication - trimming brush strips to fit length of shaft
 - McMaster Code: 74405T12
- Shaft - 3D Printed
 - Has 6 vertical 12 inch slots to hold brush strips
 - Designs to test:
 - Slide in w/ cap: check if spinning the brushes don't fly out the top

Design Intent



Image 1 (left): Shaft with bearing at the bottom and cap notches

Image 2 (middle): Shaft with bearing at the bottom and cap notches + brushes

Image 3 (right): Shaft with bearing at the bottom and cap notches + brushes + cap

Instructions to Assemble Prototype

1. 3D print shaft with brush slots
2. Cut brushes to length (12in/304mm per brush)
3. Insert brushes into brush slots
4. Secure with cap

Design Tests Descriptions and Criteria:

Test Type	Part Being Tested	What we are testing for	How the test is being performed	Results to Look At	Conclusion for next iteration
Torque Test	Motor	How strong the motor would need to be based on the speed at which we want the brushes to move.	Apply weights to a crank attached to the shaft. Continue adding weights until the shaft begins to rotate.	Given the known weight and crank length, we can compute the torque required to initialize rotation. We will need a motor of power which can supply this torque at the rotational speed required.	Once we have this starting torque, we can decide what motor we need based on the optimal rotational rate. (We will test for optimal rotation rate in the functional test).
Functional Test	Brushes	How effective is the brush structure at sweeping objects off of cylindrical objects at given rotational rates.	Stick 5 strips of sticky notes which have been cut 2cm thick and rolled into rings onto a rod. Assess how many the brush can take off after approximately 10 revolutions. Repeat the test 3 times.	Count how many of the 5 sticky notes have been removed.	At least 4/5 of the sticky notes should be removed for this to 'pass' the test. If this requirement has not been met, we need to explore different bristle diameters (how stiff the bristles are) and/or the length of bristles.
Assembly Test (Snugness of fit)	Brushes and slots in shaft	Testing for correct tolerancing between the slots in the shaft and the bases of the brush such that they fit snugly and there are no large gaps.	Place the brushes into the slots. Test for how large the gap is between the base of the brush and the slot using sheets of paper: first try placing a strip of paper through the gap. If that can fit through, fold the paper in two and try to fit this into the gap. Continue the process until you can no longer fit the paper into the gap.	Record the largest thickness of paper which fits inside the gap as a way to quantify the tolerance.	If the test yields a gap larger than 2 folds (4 layers of paper) or requires excessive force, we will need to redesign the slots in the shaft to provide a better fit.

Design Test Results and Conclusions

Test Type	Results	Conclusion for next iteration
Torque Test	Moment arm: 1.75cm Test 1: 50g attached did not rotate the design Test 2: 100g attached rotated the design Torque required to initialize rotation (using mass from test 2): $T = 0.02\text{Nm}$ 	We will require a low power motor to rotate the shaft. Although the torque seems very low, we think that with the friction of attaching it to the arms would cause it to be higher later on. This will need to be looked at when we fabricate/buy the arms to attach the shafts to in a more refined prototype.
Functional Test	Test 1: 3/5 removed Test 2: 5/5 removed Test 3: 5/5 removed 	The test has passed - on average more than 4 sticky notes were removed. The brushes are of correct stiffness to remove objects effectively. One change we will make is increasing the length of the bristles as they were smaller than expected which made it difficult to reach the sticky notes unless they were very close to the shaft itself. Key conclusions: • Brushes correct stiffness • Bristles are too small - order new brushes
Assembly Test	Test 1 (first print of shaft): There was no gap between the brush and the slot, resulting in us partially snapping the slot in order to fit the brushes in.	Test 1: Failed - brushes required excessive force causing damage to the slots (no gap) Test 2 - Passed - the brushes slid in smoothly and stayed in place without falling out - The gap can be roughly described as 1 sheet of paper thick



Picture of first shaft design



Picture showing that the brushes needed to be forced into the slots.

Changes made for test 2:

- Increased the brush slot length by 1.27 inches to allow for easier insertion.
- Smoothed the slot edges to reduce friction during assembly.
- Goal: achieve a snug fit without excessive force or looseness

Key conclusions:

- No gap between the brush and shaft is too tight and makes assembly difficult and unreliable
- Adding a small amount of clearance makes assembly much easier while still keeping the brushes secure
- Increasing the slot length and smoothing the edges helped reduce friction during insertion
- The second design achieves a good balance between ease of assembly and retention

The current shaft design is good to move forward. There will likely be adjustments needed later at the top and bottom:

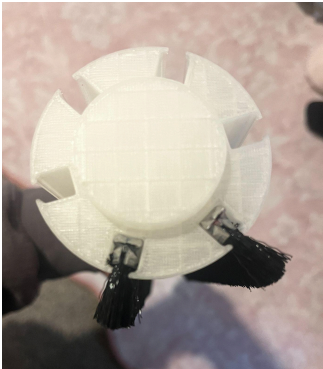
- The top will depend on how we finalize the gear/motor design
- The bottom will depend on the arm connection details

Test 2 (second print of shaft):

The updated design had enough clearance for the brushes to slide in while still being tight enough to stay in place without falling out



Picture showing the brushes installed in the shaft, evenly aligned and seated properly within the slots



Picture showing the slot openings and how the brushes are held in place



Picture showing the cap used to secure the brushes in place, preventing them from shifting or coming out.

Success Criteria

A spinning mechanism that effectively removes objects with ample power and stability

- The brush should be able to remove 80% of sticky notes cut into strips of 2cm after 20 revolutions when rotating at its operational speed. This is a high priority.
- A brush mechanism which is securely suspended by two arms. The mechanism should be able to carry out 50 revolutions with any deformations below 1cm.
- The brushes should be securely fit in the shaft and should not come loose after active use. We should feel no noticeable change in how snug the brush fits into the slot after 50 revolutions of the brush actively brushing on a surface.
- The brushes should be easily and quickly replaceable: we should be able to remove and replace an individual brush in under 30 seconds without excessive effort.

Exhibit day test: For exhibit day we think the most important test to show will be the functional test. We will have something closer to a grape vine with attachments that are more like actual lantern flies attached to show the effectiveness of our design. Our machine should remove as many of the fake bugs as possible to show that it is successful.